Sri Chaitanya IIT Academy.,India.
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A right Choice for the Real Aspirant
ICON Central Office - Madhapur - Hyderabad

SEC: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT | JEE-MAIN | Date: 22-01-2026
Time: 09.00Am to 12.00Pm | GTM-17 | Max. Marks: 300

KEY SHEET

MATHEMATICS

1	**1**	2	**1**	3	**2**	4	**2**	5	**3**
6	**4**	7	**2**	8	**3**	9	**4**	10	**3**
11	**2**	12	**1**	13	**1**	14	**1**	15	**3**
16	**2**	17	**4**	18	**3**	19	**1**	20	**4**
21	**219**	22	**833**	23	**3**	24	**2**	25	**9**

PHYSICS

26	**2**	27	**4**	28	**3**	29	**2**	30	**4**
31	**1**	32	**2**	33	**2**	34	**4**	35	**2**
36	**3**	37	**2**	38	**2**	39	**2**	40	**2**
41	**1**	42	**3**	43	**1**	44	**1**	45	**4**
46	**1**	47	**3**	48	**3**	49	**3**	50	**8**

CHEMISTRY

51	**3**	52	**4**	53	**2**	54	**2**	55	**1**
56	**3**	57	**3**	58	**2**	59	**1**	60	**4**
61	**3**	62	**2**	63	**4**	64	**4**	65	**2**
66	**4**	67	**3**	68	**4**	69	**4**	70	**2**
71	**150**	72	**6**	73	**15**	74	**8**	75	**6**

For More Join: @IITJEE_Advanced

SOLUTION

MATHEMATICS

1. $\dfrac{(2,3\sin\theta)}{(1,-2\sin\theta)}$ real part $=0$ $\Rightarrow$ $2-6\sin^2\theta=0$ $\sin^2\theta=\dfrac{1}{3}$

2. $x^2+2x+3=0$ $\Rightarrow$ $\Delta=4-12=-8<0$

$\Rightarrow$ both roots are common for the given equations

$\therefore$ $\dfrac{a}{1}=\dfrac{b}{2}=\dfrac{c}{3}\Rightarrow a:b:c=1:2:3$

3. $\dfrac{k+1}{k}=\dfrac{8}{k+3}\neq\dfrac{4k}{3k-1}$ $k=3$

4. Four digit numbers will start from 6, 7, 8

$3\times4\times3\times2=72$, Five digit numbers $=5!=120$

Total number of integers $=192$

5. $\dfrac{7}{9}[0.9+0.99+0.999+.....20\ terms]$

$=\dfrac{7}{9}\left[(1-0.1)+(1-0.01)+(1-0.001)+.....20\ terms\right]$

$=\dfrac{7}{9}\left[20-\left\{\dfrac{1}{10}+\left(\dfrac{1}{10}\right)^2+\left(\dfrac{1}{10}\right)^3+.....20\ terms\right\}\right]$

$=\dfrac{7}{9}\left[\dfrac{179+10^{-120}}{9}\right]=\dfrac{7}{81}\left[179+10^{-20}\right]$

6. $\lim\limits_{x\to0}\dfrac{\sin\left(\pi-\pi\sin^2x\right)}{x^2}$ $\lim\limits_{x\to0}\dfrac{\sin\left(\pi\sin^2x\right)}{\pi\sin^2x}\times\dfrac{\pi\sin^2x}{x^2}=1\times\pi\times1=\pi$

7. Conceptual

8. $f(-1)=0$ and $f(2)=0$, Solving above equations we get $\alpha=2,\beta=-\dfrac{1}{2}$.

9. $\int\limits_a^b f(x)dx=\int\limits_a^b f(a+b-x)dx$

10. $I=-\int\dfrac{-\left(\dfrac{2}{x^3}+\dfrac{5}{x^6}\right)}{\left(1+\dfrac{1}{x^2}+\dfrac{1}{x^5}\right)^3}=\dfrac{1}{2\left(1+\dfrac{1}{x^2}+\dfrac{1}{x^5}\right)}+C=\dfrac{x^{10}}{2\left(x^5+x^3+1\right)}+C$

11. Mid point $QR=\left(\dfrac{13}{2},1\right)$, Slope $PS=-\dfrac{2}{9}$

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Eq. of the line is $y+1=\frac{-2}{9}(x-1) \Rightarrow 2x+9y+7=0$

12. Conceptual

13. Foci of the given ellipse are $(\pm\sqrt{7},0)$

Equation of the circle is $x^2+(y-3)^2=7+9 \Rightarrow x^2+y^2-6y-7=0$

14. $l^2-m^2-n^2=0 \quad l=-(m+n)$

$(m+n)^2-m^2-n^2=0 \quad 2mn=0$

$m=0(or)\ n=0 \quad If\ m=0\ \ l=-(0+n) \quad l=-n$

$l:m:n=-n:0:n=-1:0:1$

Similarly if $n=0$ then

$l:m:n=-1:1:0\ \cos\theta=\frac{1+0+0}{\sqrt{2}\sqrt{2}} \Rightarrow \theta=\frac{\pi}{3}$

15. $\overline{AB}+\overline{AC}=2\overline{AD},\ \ \overline{AD}=\frac{1}{2}(\overline{AB}+\overline{AC})=\frac{1}{2}(8\hat{i}-2\hat{j}+8\hat{k})=4\hat{i}-\hat{j}+4\hat{k}$

$|\overline{AD}|=\sqrt{16+1+16}=\sqrt{33}$

16. $(2+\sin x)\frac{dy}{dx}+(y+1)\cos x=0 \quad \frac{2+\sin x}{\cos x}+(y+1)\frac{dx}{dy}=0$

$\ln(y+1)+\ln(2+t)=\ln c$

$(y+1)+(2+\sin x)=c\ \ at\ \ x=0 \Rightarrow C=4$

$(y+1)+(2+\sin x)=4\ \ at\ \ x=\frac{\pi}{2} \Rightarrow y=\frac{1}{3}$

17. $f_4(x)-f_6(x)=\frac{1}{4}\left[1-2\sin^2 x\cos^2 x\right]-\frac{1}{6}\left[1-3\sin^2 x\cos^2 x\right] \quad =\frac{1}{4}-\frac{1}{6}=\frac{1}{12}$

18. $\Rightarrow \quad \left(1+\cos\frac{\pi}{8}\right)\left(1+\cos\frac{3\pi}{8}\right)\left(1-\cos\frac{3\pi}{8}\right)\left(1-\cos\frac{\pi}{8}\right)$

$\Rightarrow \quad \left(1-\cos^2\frac{\pi}{8}\right)\left(1-\cos^2\frac{3\pi}{8}\right) \Rightarrow \quad \sin^2\frac{\pi}{8}\sin^2\frac{3\pi}{8}=\left(\sin\frac{\pi}{8}.\sin\frac{3\pi}{8}\right)^2$

$\Rightarrow \quad \left(\frac{2}{2}\sin\frac{\pi}{8}f\cos\frac{\pi}{8}\right)^2=\left(\frac{1}{2}\frac{\sin\pi}{4}\right)^2=\left(\frac{1}{2}.\frac{1}{\sqrt{2}}\right)^2=\frac{1}{8}fnf\ .$

19. $\sqrt{x+1}-\sqrt{x-1}=\sqrt{4x-1}$

$SBS:(x+1)+(x-1)-2\sqrt{x+1}\sqrt{x-1}=4x-1\ =-2\sqrt{x+1}\sqrt{x-1}=2x-1$

$SBS:4(x+1)(x-1)=4x^2+1-4x$

$\Rightarrow \quad (4x^2-4)=4x^2-4x+1 \quad \Rightarrow \quad 4x=5 \Rightarrow x=\frac{5}{4}$ (Rejected)

20. Equating coefficients of x^3 and x^4 is zero $\frac{-32\times17}{3}+17a-b=0$

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$16\times15-32a+3b=0$ solving $(a,b)=\left(16,\frac{272}{3}\right)$

21. $n(A\times B)=2\times4=8$

${}^8C_3+{}^8C_4+.....+{}^8C_8=2^8-\left[{}^8C_0+{}^8C_1+{}^8C_2\right]$

$=256-(1+8+28)=256-37=219$

22. Variance of first 'n' even natural numbers is equal to $\frac{n^2-1}{3}$ $=\frac{50^2-1}{3}=833$

23. $C_1(2,3);r_1=\sqrt{4+9+12}=5$

And $C_2(-3,-9),r_2=\sqrt{9+81-26}=8$

$C_1C_2=\sqrt{25+144}=13$

$C_1C_2=r_1+r_2$ touching externally.

24. For $f(x)$ to be continuous $2k=3m+2$

$2k-3m=2$ ..... (i)

For $f(x)$ to be differentiable $\frac{k}{4}=m$ $k=4m$

From (i), $8m-3m=2\Rightarrow 5m=2$ $\therefore$ $k+m=\frac{2}{5}+\frac{8}{5}=\frac{10}{5}=2$

25. $\vec{b}_1=-4\hat{i}+3\hat{j}+2\hat{k},\vec{b}_2=-4\hat{i}+\hat{j}+\hat{k},\vec{a}_1=-3\hat{i}+6\hat{j},\vec{a}_2=-2\hat{i}+7\hat{k}$

$$\left|\vec{b}_1\times\vec{b}_2\right|=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\-4&3&2\\-4&1&1\end{vmatrix}=\left|\hat{i}(3-2)-\hat{j}(-4+8)+\hat{k}(-4+12)\right|=\sqrt{1+16+64}=9$$

$\vec{a}_2-\vec{a}_1=\hat{i}-6\hat{j}+7\hat{k}$

$\left[\vec{a}_2-\vec{a}_1\ \ \vec{b}_1\ \ \vec{b}_2\right]=(\vec{a}_2-\vec{a}_1).(\vec{b}_1\times\vec{b}_2)$ $=1+24+56=81$

$\therefore$ Distance between skew lines $=\left|\frac{\left[\vec{a}_2-\vec{a}_1\ \ \vec{b}_1\ \ \vec{b}_2\right]}{\left|\vec{b}_1\times\vec{b}_2\right|}\right|=\left|\frac{81}{9}\right|=9$

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PHYSICS

26. Emf per unit temperature difference [S] $= \left[ML^2T^{-3}I^{-1}K^{-1}\right]$

Electrical conductivity $[\sigma] = \left[M^{-1}L^3T^3I^2\right]$

Thermal conductivity $[K] = \left[M^1L^1T^{-3}K^{-1}\right]$

$$[Z] = \frac{S^2\sigma}{K} = \frac{\left[ML^2T^{-3}I^{-1}K^{-1}\right]^2\left[M^{-1}L^{-3}T^3I^2\right]}{\left[M^1L^1T^{-3}K^{-1}\right]}$$

$$= \frac{\left[M^1L^1T^{-3}K^{-2}\right]}{\left[M^1L^1T^{-3}K^{-1}\right]} \quad \therefore \ [Z] = \left[M^0L^0T^0I^0K^{-1}\right]$$

27. Let after 't' time both ball and bullet hit the ground.

Then, $ft = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2\times 5}{10}} = 1\text{sec}.$

After collision let V_{ball} be velocity of ball and V_{bullet} be velocity of bullet.

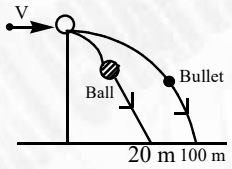
So, $20 = V_{ball} \times 1 \Rightarrow V_{ball} = 20\ m/s$

$100 = V_{bullet} \times 1 \Rightarrow V_{bullet} = 100\ m/s$

By law of conservation of momentum

$0.01V = 0.01\times 100 + 0.2\times 20 \quad \Rightarrow \quad 0.01V = 1+4$

$$\Rightarrow \quad V = \frac{5}{0.01} = 500\ m/s.$$



28. When a disc rolls, it has two types of motion i.e, translational and rotational. Therefore there are two types of angualr fmomentum and the total angular momentum is the vector sum of these two. $\vec{L} = \vec{L}_T + \vec{L}_R$

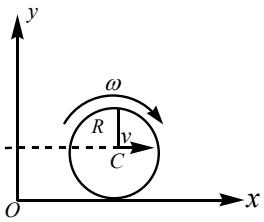
L_T = angular momentum due to translational motion.

L_R = angualr momentum due to rotational motion about C.M.

$L = MV \times R + I_{cm}\omega$

$I_{cm} = M.I.$ about centre of mass C and $V = R\omega$

$$= M(R\omega)R + \frac{1}{2}MR^2\omega = \frac{3}{2}MR^2\omega$$



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29. Pressure will be same at the same horizontal level.

Therefore, $P_A = P_B$

$\Rightarrow \quad P_0 + \rho_k g \times 0.1 + \rho_w g \times (h_1 - 0.1) = \rho_w g h_2 + \rho_0$

$\Rightarrow \quad 80 + 1000(h_1 - 0.1) = 1000 h_2$

$\Rightarrow 80 + 1000\, h_1 - 100 = 1000\, h_2$

$\Rightarrow \quad h_1 - h_2 = \dfrac{20}{1000}$

$\Rightarrow \quad h_1 - h_2 = 0.02 \quad \ldots\ldots\ldots\ldots$ (i)

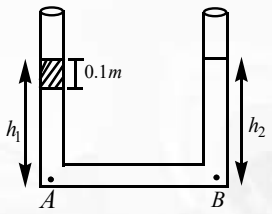
Also, $h_1 - 0.1 + h_2 = 2 \times 0.29$

$\Rightarrow \quad h_1 + h_2 = 0.68 \quad \ldots\ldots\ldots\ldots$ (ii)

Solving (i) & (ii), we get

$2h_1 = 0.70 \quad \Rightarrow \quad h_1 = 0.35\ m$

And $h_2 = 0.33\ m$ So, $\dfrac{h_1}{h_2} = \dfrac{35}{33}$

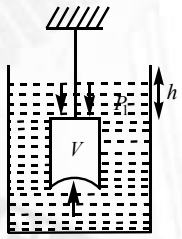


30. From Archimedes principle

Upthrust $= wt$. of fluid displaced

Now, $F_{bottom} - F_{top} = V\rho g \qquad = P_1 \times A + V\rho g$

$= (h\rho g) \times (\pi R^2) + V\rho g \quad = \rho g \left[\pi R^2 h + V\right]$



31. For cyclic process:

$Q_{cyclic} = W_{AB} + W_{BC} + W_{CA} = 10J + 0 + W_{CA} = 5J \quad \therefore \quad W_{CA} = -5J$

32. From given $P-T$ diagram,

For process JK : Isobaric $W = nR(3T_0 - T_0) = 2RT_0$

And $\Delta u = nc_v \Delta T = \dfrac{3R}{2} \times 2T_0 = 3RT_0$

For process KL : Isothermal $W = -nR(3T_0)\ln 2$

$W = -3RT_0 \ln 2$

$\Delta u = 0$ and $Q = -3RT_0 \ln 2$

For process LM : Isobaric

$W = nR(T_0 - 3T_0) = -2RT_0$

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$\Delta u = nc_v\Delta T = n\times\frac{3}{2}R\times(-2T_0) = -3RT_0$

For process MJ :Isothermal

$W = nRT\ln 2 = RT_0\ln 2$

$\Delta u = 0$

And work done in the complete cyclic process,

$W_{net} = RT_0\ln 2 + 2RT_0 - 3RT_0\ln 2 - 2RT_0$

$= -2RT_0\ln$

So correct option is (b)

33. Gravitational pull of the mass 'M' present in the sphere of radius 'r'. Provide the required centripetal force of particle of mass 'm' to revolved in a circular path.

$$\frac{mv^2}{r} = \frac{GMm}{r^2} \rightarrow \frac{1}{2}mv^2 = \frac{GMm}{2r} \Rightarrow K = \frac{GMm}{2r}$$

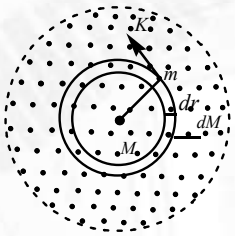
$$\therefore \qquad M = \frac{2Kr}{Gm}$$

Differentiating the above equation w.r.t 'r' we get

$$\frac{dM}{dr} = \frac{2K}{Gm} \quad \text{Or} \quad dM = \frac{2K}{Gm}dr$$

$$\therefore \qquad 4\pi r^2 dr\,\rho = \frac{2K}{Gm}dr \Rightarrow \rho = \frac{K}{2\pi r^2 mG}$$

$$\therefore \qquad \frac{\rho}{m} = \frac{K}{2\pi r^2 m^2 G} \quad or \quad \frac{\rho(r)}{m} = \frac{K}{2\pi r^2 m^2 G}$$



34. The equation for the S.H.M. $x = a\sin\omega t$

$$\Rightarrow \qquad x = a\sin\left(\frac{2\pi}{T}\right)\times t = 1\sin\left(\frac{2\pi}{8}\right)t = \sin\frac{\pi}{4}t$$

$$\left(\because \quad a = 1\,cm, T = 8s \ \ from\ \ graph\right)$$

Or, velocity, $v = \frac{dx}{dt} = \frac{d}{dt}\left[\sin\left(\frac{\pi}{4}\right)t\right] = \frac{\pi}{4}\cos\left(\frac{\pi}{4}\right)t$

$$\therefore \qquad \text{Acceleration } a = \frac{d^2x}{dt^2} = -\left(\frac{\pi}{4}\right)^2\sin\left(\frac{\pi}{4}\right)t$$

At $t = \frac{4}{3}s$ acceleration

$$a = \frac{d^2x}{dt^2} = -\left(\frac{\pi}{4}\right)^2\sin\frac{\pi}{4}\times\frac{4}{3} = \frac{-\pi^2}{16}\sin\frac{\pi}{3} \qquad = \frac{-\sqrt{3}\pi^2}{32}cm/s^2$$

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35. From lens maker's formula, focal length of convex lens (f_1)

$$\frac{1}{f_1}=(\mu-1)\left[\frac{1}{R_1}-\frac{1}{R_2}\right]=(1.5-1)\left[\frac{1}{20}-\left(\frac{1}{-20}\right)\right]\frac{1}{f_1}=\frac{1}{20} \quad \therefore \quad f_1=+20\ cm$$

Similarly, focal length of concave lens (f_2)

$$\frac{1}{f_2}=(\mu-1)\left[\frac{1}{R_1}-\frac{1}{R_2}\right] \Rightarrow \frac{1}{f_2}=(1.5-1)\left[-\frac{1}{20}-\frac{1}{20}\right]=\frac{1}{-20} \quad \therefore \quad f_2=-20\ cm$$

Now for lens L_1 $\frac{1}{v}-\frac{1}{u}=\frac{1}{f}\Rightarrow\frac{1}{v}=\frac{1}{f}+\frac{1}{u} \Rightarrow \frac{1}{v}=\frac{1}{20}+\frac{1}{-10} \therefore v=-20\ cm$

$\therefore$ Magnification $M_1=\frac{v}{u}=\frac{-20}{-10}=2$

Again for lens L_2 $u=-20-10=-30\ cm, f_2=-20\ cm$

$$\frac{1}{v}=\frac{1}{f}+\frac{1}{u}=\frac{1}{-20}+\frac{1}{-30}\Rightarrow \frac{1}{v}=\frac{-5}{60} \therefore v=-12\ cm$$

So, magnification, $m_2=\frac{v}{u}=\frac{-12}{-30}=\frac{2}{5}$ $\therefore$ Net magnification,

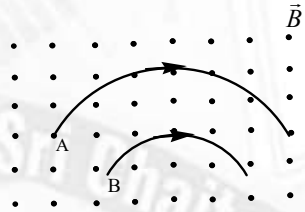
$$m=m_1m_2=2\times\frac{2}{5}=\frac{4}{5}=0.8$$

36. $E_1=\frac{1}{4\pi\in_0}.\frac{Q}{R^2}; \quad E_2=\frac{1}{4\pi\in_0},\frac{2Q}{R^2}; E_3=\frac{1}{4\pi\in_0},\frac{Q/2}{R^2}$. Clearly $E_2>E_1>E_3$

37. When a charged particle is moving normal to the magnetic field then a force acts on it which behaves as a centripetal force and moves the particle in circualr motion.

$$\therefore \quad F=qvB=\frac{mv^2}{r}\Rightarrow r=\frac{mv^2}{qB}$$

If q and B are same for both then, $r\propto mv$



Since $r_A>r_B \Rightarrow m_Av_A>m_Bv_B$

38. As resistance of wire, $R=\int\rho\frac{l}{A}$

$$R_{Fe}=\frac{\rho F_e\times l_{Fe}}{A_{Fe}}=\frac{10^{-7}\times50\times10^{-3}}{4\times10^{-6}}=\frac{25}{2}\times10^{-4}$$

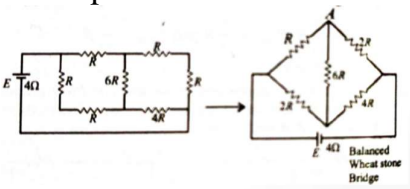
$$R_{Al}=\frac{\rho_{Al}\times l_{Al}}{A_{Al}}=\frac{2.7\times10^{-8}\times50\times10^{-3}}{(49-4)\times10^{-6}}=\frac{2.7\times50}{45}\times10^{-5}$$

As potential difference across both resistors is same, so they are in parallel combination.

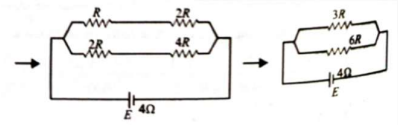
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$$\therefore \quad R_{PQ} = \frac{R_{Fe} \times R_{Al}}{R_{Fe} + R_{Al}} = \frac{12.5 \times 10^{-4} \times 0.3 \times 10^{-4}}{12.8 \times 10^{-4}} = \frac{1875}{64} \mu\Omega$$

39. The equivalent circuits are shown in the figure.



The circuit represents balacned Wheatstone Bridge. Hence no current will flow across $6R\Omega$ resistance.



$$\frac{1}{R_{eq}} = \frac{1}{3R} + \frac{1}{6R}, \Rightarrow R_{eq} = \frac{(3R)(6R)}{(3R)+(6R)} = 2R$$

For maximum power, $R_{external} = R_{\text{int}ernal}$

$2R = 4\Omega \ \therefore \ R = 2\Omega$

40. In thel given question,

Current flowing through the wire, $I = 1A$

Speed of the frame, $v = 10\ ms^{-1}$

Side of square loop, $l = 10\ cm$

Distance of square frame from current carrying wires $x = 10\ cm$.

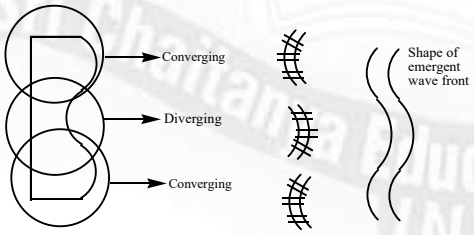
We have to find, e.m.f induced $e = ?$

According to Biot-Savart's law

$$B = \frac{\mu_0}{4\pi}\frac{Idl\sin\theta}{x^2} = \frac{4\pi \times 10^{-7}}{4\pi} \times \frac{1 \times 10^{-1}}{\left(10^{-1}\right)^2} = 10^{-6}$$

Induced e.m.f. $e = Blv = 10^{-6} \times 10^{-1} \times 10 = 1\mu v$

41. Clearly middle part of glass is diverging and upper and lower part are converging so correct shape of the emergent wavefront is as shown in the figure.



42. Statement I is true but Statement II is false
43. (A) false (R) is true
44. Both (A) and (R) are true and R is the correct explanation of A
45. Statement I is false but Statement II is true
46. Given : Force, $\vec{F} = \left(\alpha y\hat{i} + 2\alpha x\hat{j}\right)$ And $\alpha = -1Nm^{-1}$

We know that $dW = \vec{F}.d\vec{r} = \left(\alpha y\hat{i} + 2\alpha x\hat{j}\right).\left(dx\hat{i} + dy\hat{j}\right)$ $\therefore$ $dW = \alpha ydx + 2\alpha xdy$

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Work done

From $A \to B$ $dy = 0$, as $y = 1$ $\therefore$ $W_1 = \int_0^1 \alpha y dx = \alpha \int_0^1 dx = \alpha$

From $B \to C dx = 0$, as $x = 1$ $\therefore$ $W_2 = \int_1^{0.5} 2\alpha n dy = \int_1^{0.5} 2\alpha dy = 2\alpha(-0.5) = -\alpha$

From $C \to D$ $dy = 0$, as $y = 0.5$ $\therefore$ $W_3 = \int_1^{0.5} \alpha \times 0.5 dx = -\frac{\alpha}{4}$

From $D \to E$ $dx = 0$, as $x = 0.5$ $\therefore$ Total work done $W = W_1 + W_2 + W_3 + W_4$

$= \alpha - \alpha - \frac{\alpha}{4} - \frac{\alpha}{2} = -\frac{3\alpha}{4} = \frac{-3(-1)}{4} = 0.75J$

47. Here, $\vec{F} = -K\vec{r}$. So, force passes through origin.

So, $\tau_{origin} = 0 \Rightarrow$ angular momentum about origin will be conserved

So, $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{1}{\sqrt{2}} & \sqrt{2} & 0 \\ -\sqrt{2} & \sqrt{2} & \frac{2}{\pi} \end{vmatrix} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x & y & 0.5 \\ v_x & v_y & \frac{2}{\pi} \end{vmatrix} \Rightarrow \hat{k}\left[\frac{1}{\sqrt{2}} \times \sqrt{2} - \left(-\sqrt{2}\right) \times \sqrt{2}\right] = \hat{k}\left(xv_y - yv_x\right)$

$\Rightarrow$ $xv_y - yv_x = 3$

48. Using mirror formula for first position $u_1 = ?, v_1 = \frac{25}{3} cm, f = +10cm \left(= \frac{R}{2}\right)$

$\frac{1}{v_1} + \frac{1}{u_1} = \frac{1}{f}, \frac{3}{25} + \frac{1}{u_1} = \frac{1}{10}$ $\therefore$ $u_1 = -50\ m$

Using mirror formula for the second position $u_2 = ?, v_2 = \frac{50}{7}$ and $f = 10cm$

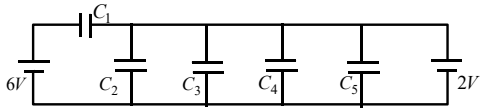
$\frac{1}{v_2} + \frac{1}{u_2} = \frac{1}{f} \Rightarrow \frac{7}{50} + \frac{1}{u_2} = \frac{1}{10} \Rightarrow \frac{1}{u_2} = \frac{1}{10} - \frac{7}{50}$

$u_2 = -25\ m$

Speed of object $= \frac{u_1 - u_2}{time} = \frac{25}{30} \times \frac{18}{5} = 3 kmh^{-1}$

49. $\phi_{cone} = \frac{q}{2 \in_0} = \frac{3q}{6 \in_0}$. So, $n = 3$

50. The circuit can be redrawn as



So, charge stroed in C_3 is given as

$Q_3 = C_3 \times 2V = 4\mu F \times 2V = 8\mu C$

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CHEMISTRY

51. $CaCO_3 + 2HCl \rightarrow CaCl_2 + CO_3 + H_2O$

Moles of $HCl = M \times V(L)$ $= 0.76 \times \frac{250}{1000} = 0.19$

Moles of $CaCO_3$ = Mass/molar mass $= \frac{1000}{100} = 10$

Therefore, HCl is the limiting reagent.

Accroding to the reaction, 2 moles of HCl

Forms $= 1$ mol of $CaCl_2$ $\therefore$ 0.19 moles of HCl will form $= \frac{0.19}{2}$

Moles of $CaCl_2$

Hence, mass of $CaCl_2$ formed $= \frac{0.19}{2} \times 111 = 10.545\ g$

52. $E_n \alpha - z^2$ 53. Statement I is correct.

Statement II is incorrect because, in sp^3d hybridization, lone pairs prefer equatorial positions, not axial, to minimize repulsion.

54. $W_{1\rightarrow 2} = -300R \ln 2$

$W_{2\rightarrow 3} = -R \times (600 - 300) = -300R$

$W_{3\rightarrow 4} = -600R \ln \frac{10}{40} = +1200R \ln 2$

$W_{4\rightarrow 1} = 0$

$W_{net} = 900R \ln 2 - 300R$

$= 300R = 660\ cal$ clockwise work -ve

55. When a non volatile solution is added is to a liquid, it lowers it's freezing point.

56. Let initially concentration of electrolytes of cathode and anode us 1 M, then $E_{cell} = E^0_{cell}$

If concentration of electrolytes of cathode and anode is 10 M, then

$$\underset{1\,atm}{H_2(g)} + \underset{1\ atm}{Cl_2(g)} \longrightarrow \underset{10\,M}{2H^+(aq)} + \underset{10\ M}{2Cl^-(aq)}$$

$E = E^0 - \frac{006}{2} \log \frac{(10)^2 \times (10)^2}{1 \times 1}$

$E = E^0 - 0.12$ $\therefore$ E decrease by 0.12 V

57. $r = k[A]^x [B]^y$

From exp $1Q2x = 1$

From exp $2Q3y = 0$

$r = k[A][B]^0$ $t_{1/2} = \frac{\ln 2}{k}$

$6.93 \times 10^{-6} = k \times 0.01$

$k = 6.93 \times 10^{-4}$

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$$t_{1/2} = \frac{0.693}{6.93\times10^{-4}} = 1000 \text{ sec}$$

$$t = 50 \text{ min} = 3000 \text{ sec}$$

$$[A]_t = \frac{[A]_0}{2^3} = \frac{0.5}{8} = \frac{1}{16} M$$

$$r = 6.93\times10^{-4} \times \frac{1}{16} M \text{ sec}^{-1}$$

58.

List – I (Family)	List – II (Symbol of Element)
Pnictogen	Mc (Moscovium)
Chalcogen	Lv (Livermorium)
Halogen	Ts (Tennesine)
Noble gas	Og (Oganesson)

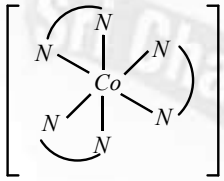
59. Due to inert pair effect, the relative stability of +1 oxidation state progressively increases for heavier elements. ∴ Stability order is: $A\ell^{1+} < Ga^{1+} < In^{1+} < T\ell^{+1}$

60. Statement I : H_2Se is less acidic than H_2Te. As we move down Group 16 in the periodic table, the bond strength decreases, making the hydrogen atoms easier to dissociate and making the acid stronger. Therefore, H_2Te is more acidic than H_2Se.
Statement II: The bond dissociation enthalpy of H_2Se is higher than H_2Te due to stronger Se-H bonds compared to Te-H bonds, so H_2Se has a higher bond enthalpy for dissociation.

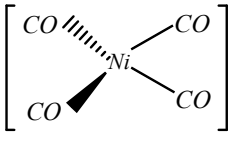
61. $$NaCl + K_2Cr_2O_7 + H_2SO_{4(conc.)} \longrightarrow \underset{(Deep\ red)}{CrO_2Cl_2}\uparrow$$

$$CrO_2Cl_2 + NaOH \longrightarrow \underset{Yellow}{Na_2CrO_4} + NaCl + H_2O$$

$$CrO_2Cl_2 + H_2O \longrightarrow \underset{Yellow}{H_2CrO_4} + HCl$$

62. P)  ⇒ coordination number = 6

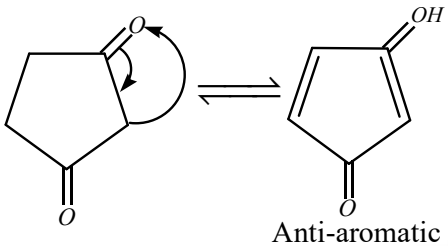
Q) $EDTA^{4\Theta}$ = Hexadentate ligand ⇒ coordination number = 6

R)  ⇒ coordination number = 4

S) Two ligands ⇒ coordination number = 2

63.

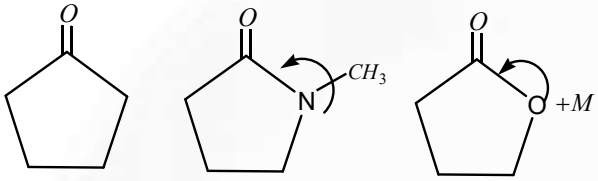
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Anti-aromatic

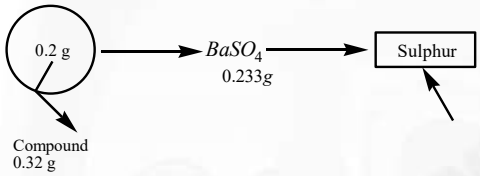
Enol form is anti-aromatic unstable, thus it does not undergo tautomerism.
Hence 100% keto content.

64.



i ii iii

65.

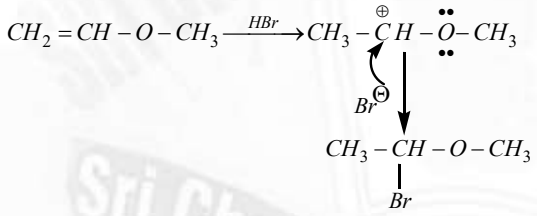


Sulphur in $BaSO_4$ sample $= \frac{32}{137+32+64} \times 0.233 = 0.032g$

% of S in organic compound $= \frac{0.032}{0.32} \times 100 = 10\%$

66. $Fe_4\left[Fe(CN)_6\right]_3$ (Prussian blue)

67.



68. $+M$ *of* $-\ddot{O}-$ *group*

69. At iso-electric point Concentration of cation is equal to concentration of anion. Net charge is zero. Maximumconc. Of di-pole ion (Zwitter ion) will be present.

70. (4) $Ph-\overset{O}{\overset{\|}{C}}-NH_2$ is weak base because of resonance.
(3) One to $-M$ of NO_2 and ortho effect it is weak base.
(2) Strong base lone pair is not involved in resonance.

71. $q = 0, \Delta U = w \Rightarrow \Delta U = nCv\Delta T$
$1 \times 20 \times \left[T_2 - 300\right] = -3000$
$T_2 - 300 = -150 \Rightarrow T_2 = 150K$

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72. $\Delta G^0 = -RT\ln K_{eq}$

And $K_{eq} = \frac{K_f}{K_b} = \frac{10^3}{10^2} = 10$

$\therefore \quad \Delta G = -RT\ln 10$

$= -(8.3\times 300\times 2.3)$

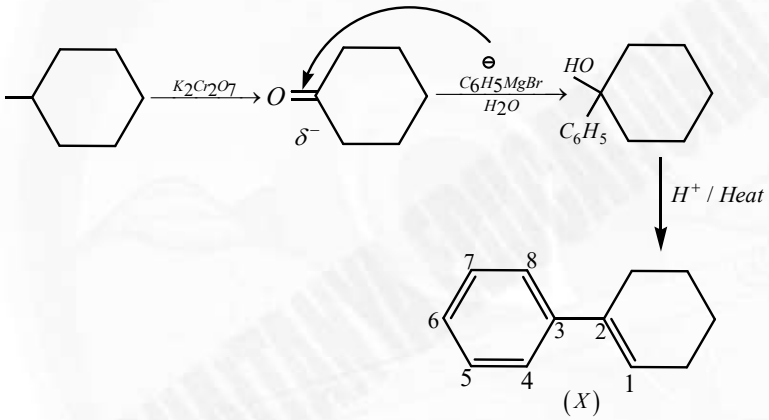
$= -5.7\ kJ\ mole^{-1} \approx 6\ kJ\ mol^{-1}$.

73. Moles of ascorbic acid $= \frac{10.2g}{176g\,mol^{-1}} = 0.058mol$

$\Delta T_f = \left(x\times 10^{-1}\right)^0 C = K_f \text{molality}$

$= 3.9\times\frac{0.058}{150}\times 1000 = 1.5^0 C = 15\times 10^{-1}\,{}^0C$

74.



(X)

75. A tri-peptide contains 3 amino acids linked by two peptide. If A, B, Care 3 different amino acids, then 6tri-peptides bonds will be formed.

1. $A-B-C$
2. $A-C-B$
3. $B-A-C$
4. $B-C-A$
5. $C-A-B$
6. $C-B-A$

Three different amino acids can be arranged by 3! In short, $3\times 2 = 6$

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